

Cutting Edge: Bispecific $\gamma\delta$ T Cell Engager Containing Heterodimeric BTN2A1 and BTN3A1 Promotes Targeted Activation of $V\gamma9V\delta2^+$ T Cells in the Presence of Costimulation by CD28 or NKG2D

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$V\gamma9V\delta2^+$ T cell–targeted immunotherapy is of interest to harness its MHC-independent cytotoxic potential against a variety of cancers. Recent studies have identified heterodimeric butyrophilin (BTN) 2A1 and BTN3A1 as the molecular entity providing “signal 1” to the $V\gamma9V\delta2$ TCR, but “signal 2” costimulatory requirements remain unclear. Using a tumor cell–free assay, we demonstrated that a BTN2A1/3A1 heterodimeric fusion protein activated human $V\gamma9V\delta2^+$ T cells, but only in the presence of costimulatory signal via CD28 or NK group 2 member D. Nonetheless, addition of a bispecific $\gamma\delta$ T cell engager BTN2A1/3A1-Fc-CD19scFv alone enhanced granzyme B–mediated killing of human CD19⁺ lymphoma cells when cocultured with $V\gamma9V\delta2^+$ T cells, suggesting expression of costimulatory ligand(s) on tumor cells is sufficient to satisfy the “signal 2” requirement. These results highlight the parallels of signal 1 and signal 2 requirements in $\alpha\beta$ and $\gamma\delta$ T cell activation and demonstrate the utility of heterodimeric BTNs to promote targeted activation of $\gamma\delta$ T cells. *The Journal of Immunology*, 2022, 209: 1475–1480.

$\gamma\delta$ T cells represent a minority subset (1–10%) of circulating T lymphocytes, but they play conserved roles in immune surveillance against microbial pathogens and malignant neoplasms (1). Overall, $\gamma\delta$ T cells exhibit properties of both the innate and adaptive immune systems, and their transcriptional program overlaps with the profiles of CD8⁺ T cells and NK cells (2). Target recognition by $\gamma\delta$ T cells via NK receptors (NKR), as well as TCR, has been demonstrated in a variety of experimental settings (3). In addition to their robust cytotoxic potential, the presence of tumor-infiltrating $\gamma\delta$

T cells represents a strong favorable prognostic marker for overall survival in multiple solid and hematological cancer types (4), validating the development of $\gamma\delta$ T cell–targeted therapies to promote antitumor immunity.

The majority of $\gamma\delta$ T cells in the peripheral blood of humans express a TCR composed of $V\gamma9$ and $V\delta2$ chains. In the context of antitumor immunity, $V\gamma9V\delta2^+$ T cells respond to transformed cells by sensing elevated phosphorylated nonpeptide metabolites or phosphoantigens (pAgs), such as isopentenyl pyrophosphate, produced via the mevalonate pathway of cholesterol synthesis that becomes dysregulated in certain tumor cells (5). pAg sensing by $V\gamma9V\delta2^+$ T cells is TCR dependent and requires engagement with B7-related membrane proteins butyrophilin (BTN) 2A1 and BTN3A1 on tumor cells. In tumor cells, pAg binding to BTN3A1 initiates a conformational change in its extracellular domain (ECD) (6), which facilitates association with BTN2A1 that can in turn engage with $V\gamma9V\delta2$ TCR (7–9). Consistent with this model, treatment of tumor cells with agonistic anti-BTN3A1/CD277 enhances $\gamma\delta$ T cell–mediated killing, but the enhanced effect is abrogated in the absence of BTN2A1 (10, 11).

Although the role of BTN2A1/3A1 in providing “signal 1” to engage $V\gamma9V\delta2$ TCR is characterized, it remains unclear whether additional costimulation is required that parallels the signal 1 and signal 2 requirements of $\alpha\beta$ T cell activation. T cell costimulatory receptors, including CD28, CD27, and 4-1BB, have been shown to synergize with TCR signaling in $V\gamma9V\delta2^+$ T cells to promote effector function, proliferation, and survival (12–14). In addition, activating NKRs also have the potential to costimulate $V\gamma9V\delta2^+$ T cells. NK group 2 member D (NKG2D) is constitutively expressed on $\gamma\delta$ T cells and recognizes stress-induced ligands, including MHC class I–related molecules A or B and UL16-binding protein molecules on infected or transformed cells (15). Engagement of NKG2D

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Abbreviations used in this article: BTN, butyrophilin; DNAM-1, DNAX accessory molecule-1; J76, Jurkat 76; NKG2D, NK group 2 member D; NKR, NK receptor; pAg, phosphoantigen.

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has been shown to amplify TCR signaling (16), as well as stimulate tumor-killing activity in TCR-dependent (17) and -independent (18, 19) manners in V γ 9V δ 2⁺ T cells. Similarly, activation of DNAX accessory molecule-1 (DNAM-1), another activating NKR, has demonstrated involvement in V γ 9V δ 2⁺ T cell-mediated cytotoxicity against acute myeloid leukemia and hepatocellular carcinoma cells (20, 21). The interplay and hierarchy between TCR and NKR in tumor target recognition and activation of V γ 9V δ 2⁺ T cells remain incompletely understood, because loss-of-function studies indicate varying degrees of contribution from each component depending on the tumor cell line/type interrogated (22).

In this study, we generated a bispecific $\gamma\delta$ T cell engager containing heterodimeric BTN2A1 and BTN3A1 ECDs fused via inert Fc linkers to scFv domains targeting tumor-antigen CD19 (BTN2A1/3A1-Fc-CD19scFv), to test its ability to activate V γ 9V δ 2⁺ T cells and to promote targeted killing of B cell lymphoma cells in a pAg-independent manner. Our results showcase the feasibility of recombinant BTN2A1/3A1 heterodimers in promoting targeted activation of V γ 9V δ 2⁺ T cells and demonstrate the requirement of a signal 2 via either a canonical T cell costimulatory receptor or NKR to fully activate BTN-mediated cytotoxicity in V γ 9V δ 2⁺ T cells.

Materials and Methods

Cells

Daudi, Raji, and K562 were obtained from ATCC. Jurkat 76 (J76) was obtained from Dr. Heemskerk, Leiden University Medical Center (23). Human V γ 9V δ 2⁺ T cells were expanded by zoledronate and cultured as previously described (24) from healthy donor leukopaks (StemCell Technologies).

Construct generation and protein production

The sequences for BTN2A1-Fc-CD19scFv and BTN3A1-Fc-CD19scFv were codon optimized and directionally cloned into mammalian expression vectors. Vectors were transiently cotransfected into CHOS cells or stably transfected into CHO cells, and the resulting heterodimeric fusion protein was purified using affinity chromatography. BTN2A1-Fc and BTN3A1-Fc homodimers were produced in a similar manner.

Cell-binding assays

10⁵ V γ 9V δ 2⁺ T cells or CD19⁺ lymphoma cells were incubated with Fc receptor blocking reagent (BioLegend), followed by incubation with BTN fusion proteins for 1 h in serum-free media at 4°C. Cells were washed and incubated with allophycocyanin-anti-human Fc (Jackson ImmunoResearch) in Dulbecco's PBS containing 1% BSA, 0.02% sodium azide, and 2mM EDTA for 30 min at 4°C. Cells were washed before analysis by flow cytometry. EC₅₀ was determined using built-in nonlinear regression analysis in GraphPad Prism.

In vitro V γ 9V δ 2⁺ T cell activation

Recombinant BTNs or Abs were incubated overnight at 4°C in high-binding 96-well plates (Corning) before adding 10⁵ V γ 9V δ 2⁺ T cells or J76-V γ 9V δ 2 in the presence of CD107a-allophycocyanin, Golgi-Stop, and GolgiPlug (BD Biosciences) at concentrations specified by the manufacturer. V γ 9V δ 2⁺ T cell cultures were incubated at 37°C for 4 h and stained for cell surface and intracellular markers for analysis by flow cytometry: anti-CD3 ϵ (clone UCHT1), anti-V γ 9 (clone B3), anti-IFN- γ (clone 4S.B3), anti-TNF- α (clone Mab11), all purchased from BioLegend. J76-V γ 9V δ 2 cultures were incubated overnight for analysis of CD69 (clone FN50; BioLegend) expression by flow cytometry. Data analysis was performed using FlowJo v10.8.0.

V γ 9V δ 2⁺ T and tumor cell coculture

A total of 10⁵ target cells were prebound with BTN2A1/3A1-Fc-CD19scFv or anti-CD277 (clone 20.1; Thermo Fisher) for 30 min at 4°C. A total of 10⁵ V γ 9V δ 2⁺ T cells were added to target cells and cultured for 4 or 1 h for detection of apoptotic or granzyme B⁺ tumor cells, respectively. Apoptotracker Green (BioLegend) was used for detection of apoptotic tumor cells, and granzyme activity in tumor cells was analyzed using the GranToxiLux assay kit (OncoImmunitin).

Statistical analysis

Graphing and statistical analysis were performed using GraphPad Prism. Unless noted otherwise, values plotted represent the mean triplicates, and error bars denote SD. Statistical significance (*p* value) was determined using unpaired Student *t* test. Significant *p* values are labeled with one or more asterisks, denoting **p* < 0.05, ***p* < 0.01, ****p* < 0.001, and *****p* < 0.0001.

Results and Discussion

V γ 9V δ 2⁺ T cell activation by recombinant BTNs requires a costimulatory signal via NKR or T cell costimulatory receptor

The ability of recombinant BTNs to promote degranulation and cytokine production in V γ 9V δ 2⁺ T cells was first assessed using homodimeric BTN2A1-Fc and/or BTN3A1-Fc chimera proteins. Although both anti-CD3 and anti-pan-TCR $\gamma\delta$ potently activated V γ 9V δ 2⁺ T cells in vitro, neither BTN2A1, BTN3A1, nor BTN2A1+BTN3A1 combination (1:1 ratio) led to degranulation or production of cytokines IFN- γ and TNF- α (Supplemental Fig. 1A). The lack of V γ 9V δ 2⁺ T cell activation by BTN2A1 and BTN3A1 suggested that either recombinant BTNs were not in an "active" conformation to engage with TCR, or that a second costimulatory signal was needed to induce V γ 9V δ 2⁺ T cell activation. Expression of known NKRs and T cell costimulatory receptors was analyzed on both ex vivo and in vitro expanded V γ 9V δ 2⁺ T cells to identify possible costimulatory receptors that are constitutively expressed. Both sources of V γ 9V δ 2⁺ T cells expressed high levels of NKG2D and DNAM-1 but did not express significant amounts of natural cytotoxicity receptors NKp30 or NKp44 (Supplemental Fig. 1B). V γ 9V δ 2⁺ T cells also constitutively expressed T cell costimulatory receptors CD28 and CD27, but only upregulated OX40 and 4-1BB on in vitro expansion via pAg stimulation (Supplemental Fig. 1B). To test the ability of NKR or T cell costimulatory receptors to provide "signal 2," we tested agonistic anti-NKG2D and anti-CD28 alone or in combination with recombinant BTNs to activate V γ 9V δ 2⁺ T cells. While BTNs, anti-NKG2D, or anti-CD28 treatment alone led to background or low levels of degranulation and cytokine production, stimulation of V γ 9V δ 2⁺ T cells with BTN2A1+BTN3A1 in combination with anti-NKG2D or anti-CD28 resulted in increased degranulation, IFN- γ , and TNF- α production (Fig. 1A, Supplemental Fig. 2A). To further demonstrate that recombinant BTNs activated V γ 9V δ 2⁺ T cells via TCR activation, we generated a T cell line expressing $\gamma\delta$ TCR (TEG) by introducing V γ 9 and V δ 2 TCR chains in J76 cells that lack endogenous TCR expression (23, 25). Although the parental J76 did not express any components of the TCR complex, J76-V γ 9V δ 2⁺ expressed TCRV γ 9, TCRV δ 2, and CD3 on the cell surface (Supplemental Fig. 2B). Consistent with primary V γ 9V δ 2⁺ T cell activation, BTN2A1+BTN3A1 activated J76-V γ 9V δ 2⁺ TEG in the presence of anti-CD28, as indicated by CD69 upregulation (Supplemental Fig. 2D). Because NKG2D is not expressed on J76-V γ 9V δ 2⁺ (Supplemental Fig. 2C), BTNs+

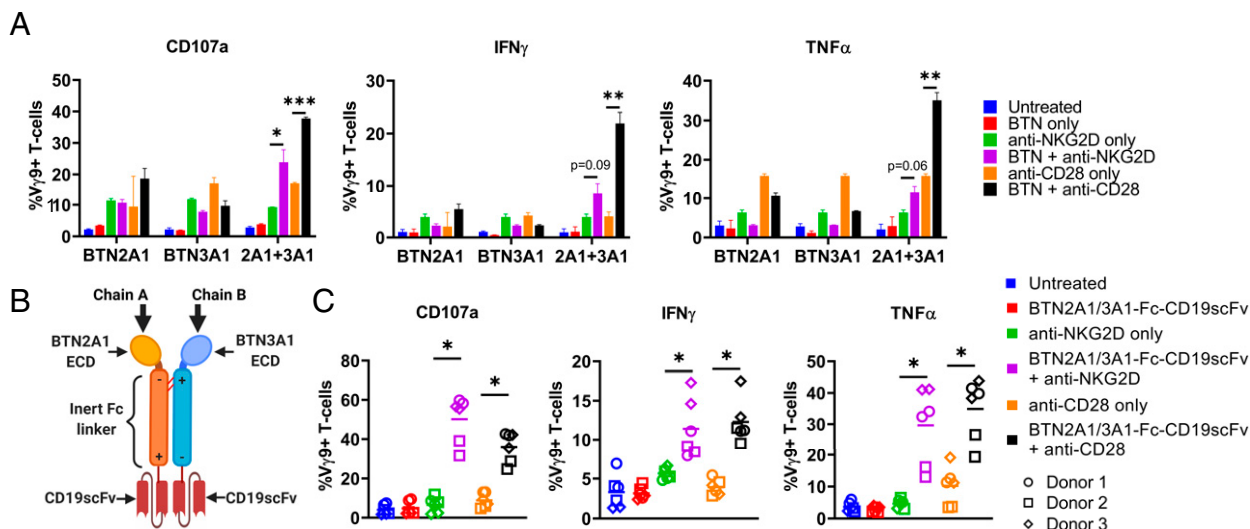


FIGURE 1. Vγ9Vδ2⁺ T cell activation by recombinant BTNs in combination with a costimulatory signal via NKR or T cell costimulatory receptor. **(A)** Vγ9Vδ2⁺ T cells were stimulated with plate-bound BTN2A1-Fc (5 μg/ml), BTN3A1-Fc (5 μg/ml), or BTN2A1+BTN3A1 (1:1 ratio, 5 μg/mL) with and without anti-NKG2D (1 μg/ml) and anti-CD28 (2.5 μg/ml). The proportion of CD3⁺Vγ9⁺ cells expressing CD107a (left panel), IFN-γ (middle panel), and TNF-α (right panel) were analyzed. Mean ± SD from three biological replicates is shown. **p* < 0.05, ***p* < 0.01, ****p* < 0.001 by Student *t* test. Data are representative of at least five independent experiments. **(B)** Schematic representation of the heterodimeric BTN2A1/3A1-Fc-CD19scFv engager construct comprising two polypeptide chains (A and B) brought together by interchain disulfide bonds and charge polarized linkers. Created with BioRender.com. **(C)** Vγ9Vδ2⁺ T cells derived from three different human donors were stimulated with plate-bound BTN fusion proteins (10 μg/ml) containing BTN2A1/BTN3A1 heterodimers with and without anti-NKG2D (1 μg/ml) or anti-CD28 (2.5 μg/ml). The proportion of CD3⁺Vγ9⁺ cells expressing CD107a (left panel), IFN-γ (middle panel), and TNF-α (right panel) was determined. Horizontal bars denote mean of two biological replicates from three different donors. **p* < 0.05 by Wilcoxon matched-pairs signed rank test. Data are representative of at least five independent experiments.

anti-NKG2D did not lead to TEG activation. Although BTN2A1 + anti-CD28 also upregulated CD69 in J76-Vγ9Vδ2⁺ TEG (Supplemental Fig. 2D), cytokine production was observed only when Vγ9Vδ2⁺ T cells were stimulated by both BTN2A1 and BTN3A1 (Fig. 1A). These results confirm the involvement of BTN2A1 and BTN3A1 in TCR-dependent activation of Vγ9Vδ2⁺ T cells and demonstrate the requirement of a “signal 2” for BTN-mediated activation of Vγ9Vδ2⁺ T cells. Furthermore, these results suggest that close proximity of plate-bound BTN2A1 and BTN3A1 homodimers was sufficient to mimic the active form of BTNs to engage with Vγ9Vδ2 TCR.

Heterodimeric BTN2A1/BTN3A1 activates Vγ9Vδ2⁺ T cells in the presence of a costimulatory signal

We generated a bispecific γδ T cell engager containing heterodimeric BTN2A1 and BTN3A1 ECDs fused via inert Fc linkers to scFv domains specific for CD19 (Fig. 1B) to test its ability to modulate Vγ9Vδ2⁺ T cells and promote killing of CD19-expressing tumor cells. The presence of BTN2A1 and BTN3A1 ECDs on the two polypeptide chains on the BTN2A1/3A1-Fc-CD19scFv molecule was confirmed by Western blot under nonreduced, reduced, and deglycosylated conditions using specific Abs (Supplemental Fig. 3A). The formation of a BTN2A1/BTN3A1 heterodimer was confirmed by a dual-binding immunoassay using capture and detection Abs that bind to the individual BTN domains (Supplemental Fig. 3B). Only the BTN2A1/3A1-Fc-CD19scFv construct containing CD19scFv, but not an unrelated scFv, bound to a CD19⁺ lymphoma cell line (Supplemental Fig. 3C, left panel). Furthermore, BTN2A1/3A1-Fc-CD19scFv (but not a control construct containing a BTN3A1 homodimer) bound to Vγ9Vδ2⁺ T cells, but not Vδ1⁺ (predominantly Vγ9⁺) or CD8⁺ T cells in PBMCs (Supplemental Fig. 3C, right panel, and Supplemental Fig. 3E). BTN2A1/3A1-Fc-CD19scFv binding to Vγ9Vδ2⁺ T cells was

partially inhibited by the presence of anti-pan-TCRγδ and completely blocked by anti-Vγ9 (Supplemental Fig. 3D), confirming the specificity of heterodimeric BTN2A1 and BTN3A1 to the Vγ9 subunit of the TCR complex.

Consistent with cell-binding specificity, stimulation of J76-Vγ9Vδ2⁺ TEG, but not parental J76, with BTN2A1/3A1-Fc-CD19scFv led to robust upregulation of CD69, but only in the presence of CD28 costimulation (Supplemental Fig. 3F). Similarly, BTN2A1/3A1-Fc-CD19scFv induced Vγ9Vδ2⁺ T cell degranulation and cytokine production in the presence of NKG2D or CD28 costimulation across multiple donor-derived in vitro expanded and naive Vγ9Vδ2⁺ T cells (Fig. 1C, Supplemental Fig. 4C). Furthermore, while anti-CD28 stimulation alone led to proliferation-naive Vδ2⁺ T cells, BTN2A1/3A1-Fc-CD19scFv in combination with anti-NKG2D enhanced proliferation of Vδ2⁺ T cells (Supplemental Fig. 4A, 4B). Although stimulation with BTN2A1 or BTN3A1 homodimers alone did not elicit activation of Vγ9Vδ2⁺ T cells (Fig. 1A), a plate-bound mixture of BTN2A1 and BTN3A1 homodimers provided costimulation-dependent activation of Vγ9Vδ2⁺ T cells similar to the BTN2A1/3A1 heterodimeric engager (Fig. 1A, Supplemental Fig. 4C, 4D). Taken together, these results suggest that activation of Vγ9Vδ2⁺ T cells requires the simultaneous presence of BTN2A1, BTN3A1, and costimulation via NKR or potentially other costimulatory receptors to fully activate the cytotoxic properties of Vγ9Vδ2⁺ T cells. The comparison of the BTN2A1/3A1 heterodimer with the mixture of plate-bound BTN2A1/3A1 homodimers raises an important question of whether the Vγ9Vδ2⁺ TCR is optimally activated by closely approximated BTN2A1/3A1 domains with or without an associated heterodimerization-driven conformational change in the ECD, or whether both BTN2A1 and 3A1 simply both need to be present within the immune synapse to provide “signal 1” to the Vγ9Vδ2⁺ TCR, and these questions should be the subject of further inquiry.

Addition of BTN2A1/3A1-Fc-CD19scFv enhances V γ 9V δ 2⁺ T cell cytotoxicity against B cell lymphoma

To evaluate the ability of BTN2A1/3A1-Fc-CD19scFv to enhance V γ 9V δ 2⁺ T cell-mediated killing of tumor cells, we cultured CD19⁺ Daudi and Raji cells (Supplemental Fig. 4E) with V γ 9V δ 2⁺ T cells in the presence of BTN2A1/3A1-Fc-CD19scFv. Addition of BTN2A1/3A1-Fc-CD19scFv to the coculture resulted in an increased proportion of apoptotic tumor cells, as evidenced by detection of translocated phosphatidylserine residues on the cell surface (Fig. 2A). Similar levels of tumor killing were induced by 1–100 μ g/ml (6.7–670 nM) BTN2A1/3A1-Fc-CD19scFv, suggesting a concentration at or less than EC₅₀ for tumor cell binding by BTN2A1/3A1-Fc-CD19scFv (Supplemental Fig. 3C) can efficiently induce cytotoxicity in V γ 9V δ 2⁺ T cells. BTN2A1/3A1-Fc-CD19scFv-mediated V γ 9V δ 2⁺ T cell cytotoxicity was additionally investigated using a cell-based fluorogenic cytotoxicity assay designed to measure granzyme B activity in live target cells after the successful transfer of granzyme B by cytotoxic lymphocytes. In agreement with tumor cell apoptosis, the proportion of tumor cells exhibiting granzyme B activity (Fig. 2B), as well as secreted levels of cytokines such as IFN- γ and TNF- α (Fig. 2C), significantly increased when BTN2A1/3A1-Fc-CD19scFv was added to V γ 9V δ 2⁺ T and Daudi or Raji cell coculture, but not to tumor cells alone, confirming that the mechanism of action for BTN2A1/3A1-Fc-CD19scFv is through enhancement of V γ 9V δ 2⁺ T cell

cytotoxicity. The frequency of granzyme B⁺ tumor cells was comparable when BTN2A1/3A1-Fc-CD19scFv or a saturating dose of agonistic anti-CD277/BTN3A1 (20.1) was added to the V γ 9V δ 2⁺ T cells and tumor coculture. This further demonstrates that recombinant heterodimeric BTN2A1/3A1 can provide an activating signal to V γ 9V δ 2 TCR in a similar fashion as endogenous BTN2A1 in combination with an active conformation of BTN3A1 induced by the 20.1 Ab. Phenotypic analysis of CD19⁺ lymphoma cell lines indicated the presence of CD28 ligands CD80 and CD86 (Supplemental Fig. 4F), suggesting the costimulatory signal needed for BTN2A1/3A1-Fc-CD19scFv activity was provided by tumor cells. To assess the ability of BTN2A1/3A1-Fc-CD19scFv to enhance V γ 9V δ 2⁺ T cell-mediated killing of tumor cells expressing NKG2D ligands, we generated a CD19-expressing K562 cell line with endogenous MHC class I-related molecules A or B on the cell surface (Supplemental Fig. 4G). Whereas the addition of BTN2A1/3A1-Fc-CD19scFv to V γ 9V δ 2⁺ T cell + parental K562 coculture did not enhance tumor cell killing, BTN2A1/3A1-Fc-CD19scFv significantly enhanced V γ 9V δ 2⁺ T cell-mediated cytotoxicity of CD19-expressing K562 cells (Fig. 2A, 2B, right panels). These results highlight the ability of BTN2A1/3A1-Fc-CD19scFv in promoting targeted killing of tumor cells expressing the select tumor Ag, as well as ligands for CD28 and/or NKG2D to deliver “signal 2” to activate V γ 9V δ 2⁺ T cells. Curiously, the maximal killing of Daudi cells occurred

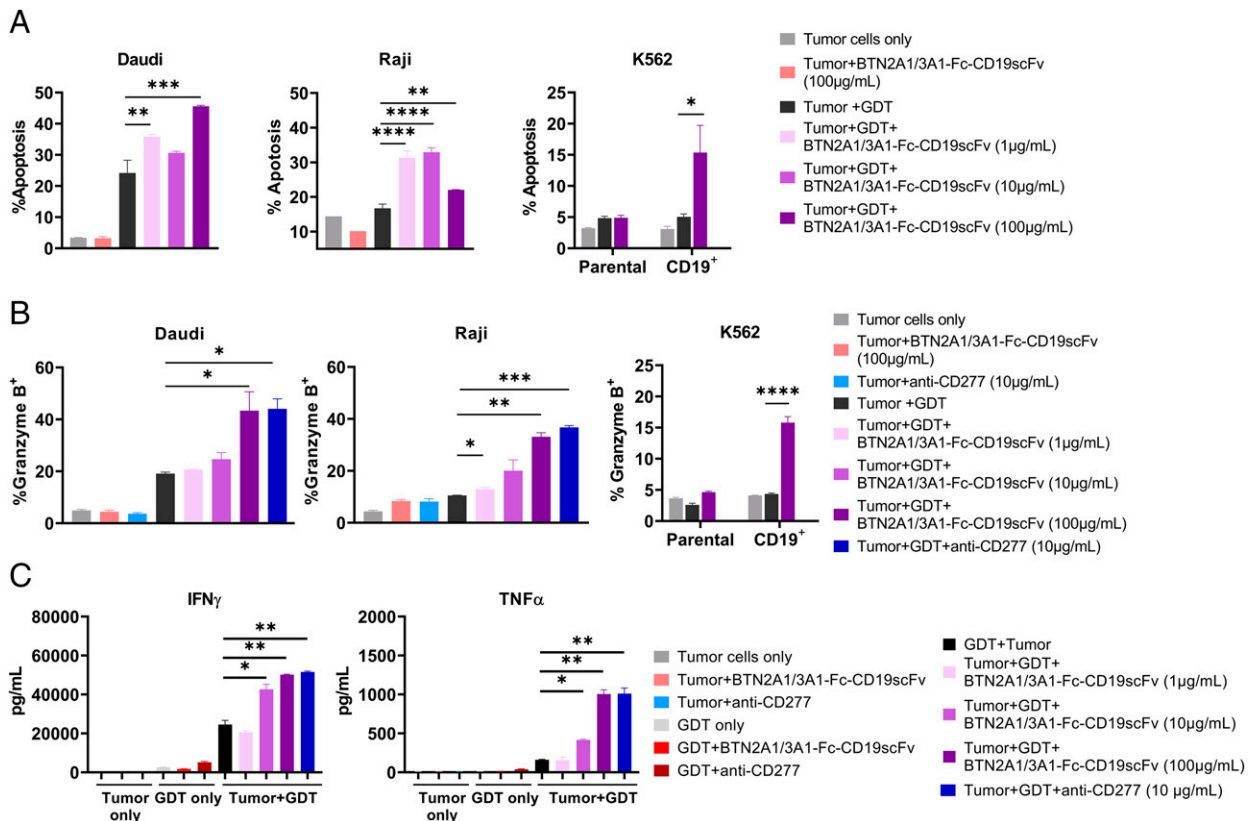


FIGURE 2. BTN2A1/3A1-Fc-CD19scFv enhanced tumor killing in vitro as a single agent. **(A)** V γ 9V δ 2⁺ T cells (GDT) were cocultured with the indicated tumor cells at 1:1 ratio in the presence of BTN2A1/3A1-Fc-CD19scFv. The proportion of apoptotic tumor cells was detected by Apotracker-Green⁺ cells in CD3⁺CD20⁺ cells (Daudi and Raji) or CD3⁺ cells (K562). Mean $\pm$ SD from three biological replicates is shown. Data are representative of at least three independent experiments from three different V γ 9V δ 2⁺ T cell donors. **(B)** V γ 9V δ 2⁺ T cells were cocultured with the indicated tumor cells at 1:1 ratio in the presence of BTN2A1/3A1-Fc-CD19scFv or anti-CD277/BTN3A1. The proportion of tumor cells with active granzyme activity was detected by flow cytometry, as determined as %Granzyme B⁺ cells in fluorescently labeled tumor cells. Mean $\pm$ SD from three biological replicates is shown. * p < 0.05, ** p < 0.01, *** p < 0.001 by Student t test. Data are representative of at least three independent experiments from three different V γ 9V δ 2⁺ T cell donors. **(C)** IFN- γ (left) and TNF- α (right) levels in supernatant from V γ 9V δ 2⁺ T and Raji coculture in (A) were quantified by human U-PLEX T-Cell Combo immunoassay (MSD). Mean $\pm$ SD from three biological replicates is shown. Data are representative of at least three independent experiments from three different V γ 9V δ 2⁺ T cell donors. * p < 0.05, ** p < 0.01, *** p < 0.001, **** p < 0.0001 by Student t test.

in the presence of 100 $\mu\text{g/ml}$ BTN2A1/3A1-Fc-CD19scFv, while the same was achieved at 1 or 10 $\mu\text{g/ml}$ for Raji cocultures (Fig. 2A). Because Raji expressed a slightly lower level of CD19 (Supplemental Fig. 4E), we speculate that when adding a high concentration of BTN2A1/3A1-Fc-CD19scFv, a higher level of unbound fusion proteins remained in the culture after the initial tumor prebinding step. These “free” BTN2A1/3A1-Fc-CD19scFv may occupy the TCRs on V γ 9V δ 2⁺ T cells and prevent the direct cell–cell interaction that is needed to mediate tumor killing. In addition, Raji expressed a higher level of CD80 (Supplemental Fig. 4F) that likely enabled stronger CD28 costimulation on V γ 9V δ 2⁺ T cells, and therefore a lower level of BTN2A1/3A1 may be needed to trigger cytotoxic functions. Nonetheless, while the dose response varied between the two lymphoma cell lines for the apoptosis readout, the proportion of granzyme B transferred into tumor cell lines was highest at 100 $\mu\text{g/ml}$ for both lymphoma cell lines. We speculate that there might also be additional mechanisms used by V γ 9V δ 2⁺ T cells to mediate cytotoxicity against lymphoma cells (i.e., via Fas or TRAIL) that can account for the differences seen in the Raji and Daudi cocultures.

Collectively, results from this study have demonstrated the feasibility of using recombinant heterodimeric BTN2A1 and BTN3A1 in a bispecific engager format to enhance antitumor activity of V γ 9V δ 2⁺ T cells. While in a tumor-free culture system we demonstrated the need for the presence of a costimulatory signal to activate V γ 9V δ 2⁺ T cells by BTNs, BTN2A1/3A1-Fc-CD19scFv as a single agent was sufficient to promote tumor killing, indicating the delivery of costimulatory signal(s) by ligands natively expressed by tumor cells. We identified NKG2D and CD28 as two costimulatory receptors for V γ 9V δ 2⁺ T cells and confirmed expression of their corresponding ligands on tumor cells. Phenotypic analysis suggests additional costimulatory receptors are present on V γ 9V δ 2⁺ T cells, including DNAM-1 and CD27 (Supplemental Fig. 1B), which likely contribute to V γ 9V δ 2⁺ T cell activation. Likewise, multiple stress-induced molecules or ligands for NKRs are likely expressed on infected or transformed cells to promote BTN-mediated killing by V γ 9V δ 2⁺ T cells. Comprehensive investigation into additional costimulatory/coinhibitory receptors on V γ 9V δ 2⁺ T cells and regulation of cell-surface expression of the corresponding ligands in different cancer cell types will be invaluable to further our understanding of V γ 9V δ 2⁺ T cell biology. In addition, these biological insights will also facilitate translation of effective $\gamma\delta$ T cell–targeted therapies and ultimately selecting cancer patients most likely to respond to this emerging class of therapy.

While Abs against the $\gamma\delta$ TCR components (i.e., CD3, TCR γ , and TCR δ) can induce robust V γ 9V δ 2⁺ T cell activation without the need for “signal 2” (Supplemental Fig. 1A) (26), the weaker TCR signal delivered by its natural ligand in the format of BTN heterodimers has the potential to provide a more targeted approach to exert cytotoxicity in cancer cells, at the same time preventing systemic activation of T cells in the clinical setting. Preliminary data comparing antitumor activity of V γ 9V δ 2⁺ T cells in the presence of different agonists suggests comparable levels of IFN- γ and TNF- α produced by engagers containing heterodimeric BTN2A1/3A1 or anti-V δ 2. More thorough understanding of the qualitative and quantitative differences in V γ 9V δ 2⁺ T cells on stimulation by different agonists is warranted, to determine the optimal TCR signaling strength necessary to promote cell proliferation and antitumor effector functions, while preventing detrimental effects such as

immunosuppression, activation-induced cell death, and anergy in V γ 9V δ 2⁺ T cells.

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Disclosures

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